

University / Building Operations Analysis

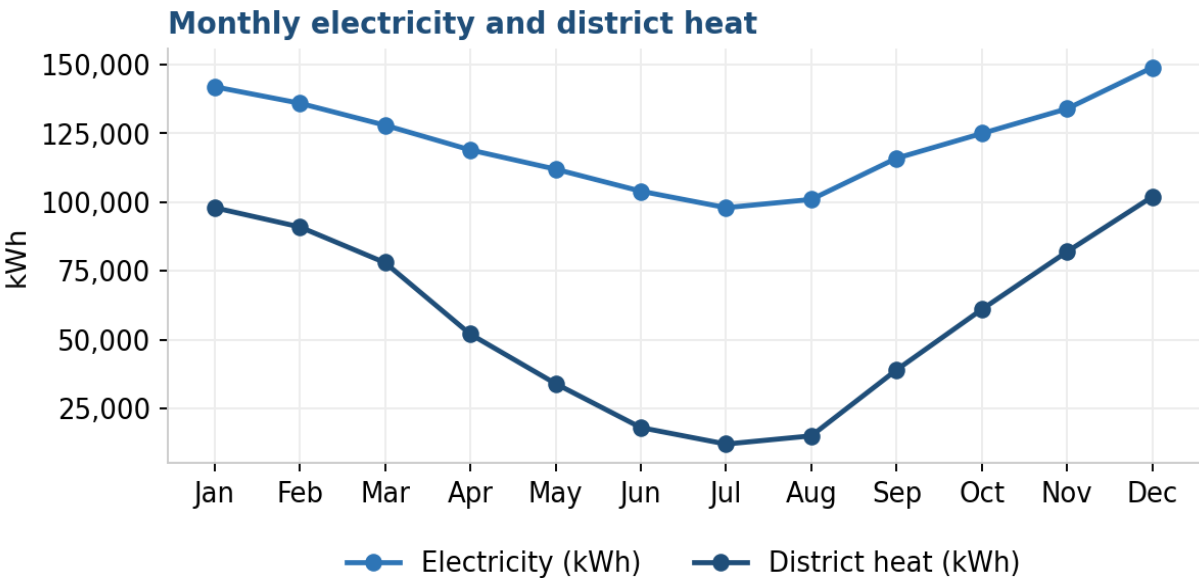
Executive summary

This synthetic case demonstrates the workflow from raw operational data to a management-ready analysis. All figures are illustrative and not real client data.

KPI	Illustrative result	Unit
Building energy	2.15	GWh
Total emissions	716.7	tCO ₂ e
Building intensity	116.0	kWh/m ²
Energy per FTE (term-time avg)	51.1	kWh/FTE-month

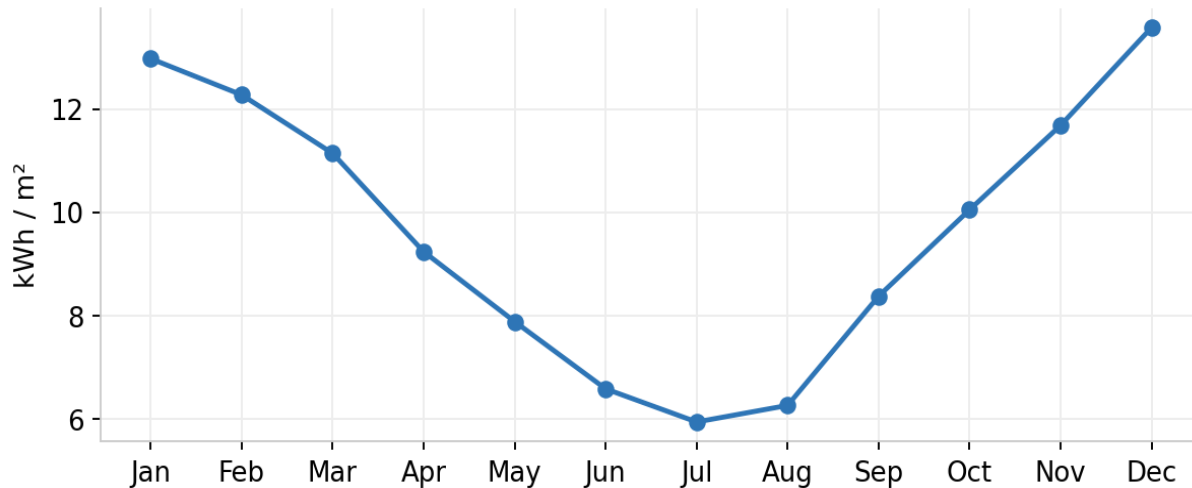
Energy & activity analysis

Building energy follows a strong seasonal, academic-calendar pattern: district heat swings 8.5x between its July trough and December peak (102,000 kWh).



Area-based intensity (kWh/m²) is stable and usable for month-to-month comparison year-round, unlike occupancy-based measures (see below).

Building intensity (kWh/m²) — stable, usable year-round

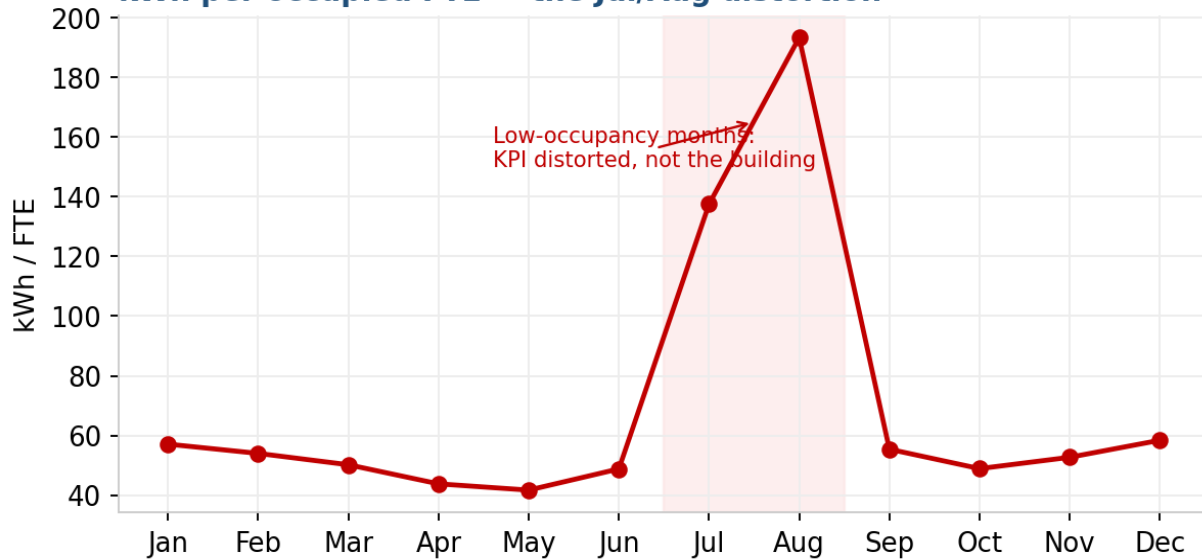


Carbon / KPI analysis

Electricity and district heat are screened as Scope 2; fleet diesel as Scope 1 in this example. Building-level meter data would enable more actionable diagnosis than campus-wide totals.

Key insight: a KPI pitfall worth flagging to any client: kWh-per-occupied-FTE spikes to 137.5 in July and 193.3 in August, 2–4x its term-time level. This is not the building using more energy, but occupancy falls faster than energy use, so the small denominator inflates the ratio. The genuine finding: Jul/Aug electricity still averages ~99,500 kWh/month with the campus nearly empty. This is a real baseload worth investigating.

kWh per occupied FTE — the Jul/Aug distortion



Data quality

Real work would reconcile meter data, define building boundaries, handle vacancies/term breaks and document methodology. This analysis auto-flags any month where occupied FTE drops below 1,000 as unsuitable for FTE-normalised comparison (2 months flagged: July, August).

Recommendations

1. Identify buildings with high baseload or heating intensity via sub-metering because campus totals hide which buildings actually drive the swing.
2. Review heating controls and schedules against the academic calendar to target the 8.5x seasonal swing.
3. Investigate the Jul/Aug baseload specifically. It is necessary to audit what is drawing ~99,500 kWh/month with the campus nearly empty (IT rooms, refrigeration, always-on HVAC).
4. Use the area-based KPI (kWh/m²), not FTE, for any month-to-month or building-to-building comparison; reserve the FTE KPI for term-time-only analysis.

Scope note: This portfolio demonstrates data preparation, screening calculations, KPI development and reporting. It is not independent GHG verification or assurance, a statutory energy audit, legal advice, or a guarantee of savings. All figures in this document are illustrative, synthetic data for demonstration purposes only.